Time allowed: 2.00 Hours | Total Marks Sections A and B: 20

SECTION – A (Marks 12)

Attempt All Questions and write answers within provided spaces on E-Sheet. تمام سوالات حل کریں اور جوابات ای شیٹ میں فراہم کردہ جگہ میں لکھیں۔

Q.1 A student is tasked with separating the mixture of lead and cadmium ions by paper chromatography. He has been provided with.

ایک طالب علم کو پیپر کرومیٹوگرافی کے ذریعے لیڈ (Lead) اور کیڈمیم (Cadmium) آئنز کے آمیزہ کو علیحدہ کرنے کا کام دیا گیا ہے۔ اسے درج اشیاء فراہم کی گئی ہیں:

i	Chromatographic paper	ii	Capillary tube
iii	Glass Jar with Lid	iv	Pencil
v	Locating agent Na_2S		

i	کرومیٹوگرافی پیپر	ii	کیپلیری ٹیوب
iii	ڈھکن کے ساتھ شیشے کا جار	iv	پنسل
v	لوکیٹنگ ایجنٹ Na_2S		

a.	Why should a pencil be used instead of an ink pen to draw the base line?	بیس لائن کھینچنے کے لیے سیاہی والے قلم کے بجائے پنسل کیوں استعمال کی جاتی ہے؟
b.	Which solvent or mixture of solvents should be used as a mobile phase?	موبائل فیز (Mobile Phase) کے طور پر کس سالوینٹ یا سالوینٹس کے مرکب کو استعمال کیا جانا چاہیے؟
c.	Which colour will be obtained for "Cd" spot on chromatogram?	کرومیٹوگرام پر "Cd" کے نشان (Spot) پر کون سا رنگ حاصل ہوگا؟
d.	Which colour will be obtained for "Pb" spot on chromatogram?	کرومیٹوگرام پر "Pb" کے نشان (Spot) کا کون سا رنگ حاصل ہوگا؟
e.	Calculate the R_f value for Pb and Cd spots by using given information. Distance travelled by solvent front=6.8cm Distance travelled by "Pb" spot=5.4cm Distance travelled by "Cd" spot=3.6cm	دی گئی معلومات کی مدد سے Pb اور Cd کے نشانات (spots) کے R_f معلوم کریں۔ دی گئی معلومات: سالوینٹ فرنٹ کا طے کیا گیا فاصلہ=6.8 cm لیڈ (Pb) کے نشان کا طے کیا گیا فاصلہ=5.4 cm کیڈمیم (Cd) کے نشان کا طے کیا گیا فاصلہ=3.6 cm

1+1 +1+1 +2

? You are tasked to identify the gases using simple experiments. Specifically damp litmus paper.

آپ کو سادہ تجربات کے ذریعے گیسوں کی شناخت کرنے کا کام دیا گیا ہے، خاص طور پر نم دار لٹمس پیپر (Damp Litmus Paper) کے ٹیسٹ استعمال کرتے ہوئے۔

a.	You have to use a damp litmus paper. What is a damp litmus paper? Explain.	آپ کو نم دار لٹمس پیپر استعمال کرنا ہے۔ نم دار لٹمس پیپر سے کیا مراد ہے؟ وضاحت کریں۔
b.	Why must the litmus paper be damp while testing ammonia?	امونیا کی جانچ کرتے وقت لٹمس پیپر کا نم دار ہونا کیوں ضروری ہے؟
c.	What colour change will be observed for litmus paper while testing in ammonia?	امونیا کی موجودگی میں لٹمس پیپر میں رنگت کی کون سی تبدیلی دیکھی جائے گی؟
d.	Write chemical equation involved when lime water reacts with carbon dioxide.	جب چونے کا پانی (Lime Water) کاربن ڈائی آکسائیڈ کے ساتھ ردعمل کرتا ہے تو اس کی کیمیائی مساوات لکھیں۔
e.	How is carbon dioxide identified using lime water?	چونے کے پانی کی مدد سے کاربن ڈائی آکسائیڈ کی شناخت کیسے کی جاتی ہے؟
f.	How is chlorine gas identified using damp litmus paper?	نم دار لٹمس پیپر کی مدد سے کلورین گیس کی شناخت کیسے کی جاتی ہے؟

1x6

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SECTION – B (Marks 8)

Q.3 A Group of students is conducting an experiment to determine the boiling point of ethanol and they are trying to assess its purity.

طلبہ کا ایک گروپ ایتھانول (Ethanol) کے نقطہ ابال (Boiling Point) کا تعین کرنے کے لیے تجربہ کر رہا ہے اور اس کے خالص پن کا جائزہ لینے کی کوشش کر رہا ہے۔

a.	What is meant by boiling point?	نقطہ ابال (Boiling Point) سے کیا مراد ہے؟
b.	What is the approximate boiling point of pure ethanol?	خالص ایتھانول کا عمومی نقطہ ابال کیا ہوتا ہے؟
c.	What observation shows that ethanol has reached its boiling point?	کون سا مشاہدہ ظاہر کرتا ہے کہ ایتھانول اپنے نقطہ ابال تک پہنچ گیا ہے؟
d.	How does impurity affect the boiling point of ethanol?	امیونٹیز ایتھانول کے نقطہ ابال (Boiling Point) کو کس طرح متاثر کرتی ہیں؟

4 Answer the following questions.

درج ذیل سوالات کے جواب دیں۔

a.	What is sublimation?	تصعید (Sublimation) کیا ہے؟
b.	How does impurity affect the melting point of a pure solid?	امیونٹیز خالص ٹھوس مادے کے نقطہ پگھلاؤ (Melting Point) کو کس طرح متاثر کرتی ہے؟
c.	What colour flame does sodium produce?	سوڈیم کس رنگ کا شعلہ پیدا کرتا ہے؟
d.	What happens to sand and salt during sublimation of mixture of Naphthalene with sand and salt?	نیفتھلین، ریت اور نمک کے آمیزہ کی تصعید (Sublimation) کے دوران ریت اور نمک کے ساتھ کیا ہوتا ہے؟

Prestige Educational Academy

Chemistry 10th PBA, EVENING

SECTION – A (Marks 12)

Attempt All Questions and write answers within provided spaces on E-Sheet. تمام سوالات حل کریں اور جوابات ای شیٹ میں فراہم کردہ جگہ میں لکھیں۔

Q.1 You are tasked to determine the exact molarity of NaOH solution volumetrically using O.1M HCl. Answer the following questions based on given experiment.

آپ کو تجربہ کے ذریعے معیاری HCl (O.1M) استعمال کرتے ہوئے والیومیٹرک طریقہ سے NaOH محلول کی درست مولیریٹی معلوم کرنے کا کام دیا گیا ہے۔ دیے گئے تجربے کی بنیاد پر درج سوالات کے جواب دیں۔

a.	What is volumetric analysis?	والیومیٹرک تجزیہ (Volumetric Analysis) کیا ہوتا ہے؟
b.	Which indicator can be used for this volumetric analysis?	اس والیومیٹرک تجزیہ کے لیے کون سا اینڈی کیٹر استعمال کیا جا سکتا ہے؟
c.	What will be the end point of this titration according to indicator mentioned in (b)?	حصہ (b) میں ذکر کیے گئے اینڈی کیٹر کے مطابق اس ٹائٹریشن کا نقطۂ اختتام (End Point) کیا ہو گا؟
d.	Write chemical reaction involved in the given titration.	دی گئی ٹائٹریشن میں شامل کیمیائی ری ایکشن (Chemical Reaction) لکھیں۔
e.	Calculate the mean volume (V_2) of HCl used taken in burette.	استعمال شدہ HCl کے حجم کی اوسط (Mean Volume) (V_2) معلوم کریں۔

S. No.	Initial reading ابتدائی ریڈنگ (cm^3)	Final reading فائنل ریڈنگ (cm^3)	Vol. of HCl used HCl کا استعمال شدہ والیوم (cm^3)
1	0	9	
2	9	18	
3	18	27	

f.	Calculate the molarity of NaOH by using given information. • Vol. of NaOH taken in flask (V_1)=10cm^3 • Molarity of HCl (M_2)=0.1M	دی گئی معلومات کی مدد سے NaOH کی مولیریٹی معلوم کریں۔ فلاسک میں لیا گیا NaOH کا حجم (V_1) = 10cm^3 HCl کی مولیریٹی (M_2) = 0.1M

1x6

Q.2 You have to separate the mixture of inks by paper chromatography. Answer the following questions based on given experiment.

آپ کو پیپر کروميٹوگرافی (Paper Chromatography) کے ذریعے سیاہی کے آمیزہ کو علیحدہ کرنا ہے۔ دیے گئے تجربے کی بنیاد پر درج سوالات کے جواب دیں۔

a.	What is meant by stationary phase in paper chromatography?	پیپر کروميٹوگرافی میں سٹیشنری فیز (Stationary Phase) سے کیا مراد ہے؟
b.	Write formula of R_f value.	R_f ویلیو کا فارمولا لکھیں۔
c.	What is meant by mobile phase in paper chromatography?	پیپر کروميٹوگرافی میں موبائل فیز (Mobile Phase) سے کیا مراد ہے؟
d.	What is chromatogram?	کروميٹوگرام (Chromatogram) کیا ہوتا ہے؟
e.	Calculate the R_f value for the spots according to given information.	دی گئی معلومات کے مطابق نشانات (Spots) کی R_f ویلیو معلوم کریں۔
	Distance travelled by solvent = 8cm Distance travelled by blue spot = 3cm Distance travelled by red spot = 5cm	سالوینٹ (Solvent) کا طے کیا گیا فاصلہ = 8cm نیلی روشنائی (Blue spot) کا طے کیا گیا فاصلہ = 3cm سرخ روشنائی (Red spot) کا طے کیا گیا فاصلہ = 5cm

1x4 +2

KHATTAK ACADEMY WAH CANTT
Imagitor

SECTION – B (Marks 8)

Q.3 A Group of students were tasked to identify the metal ions by flame test. Answer the following questions based on given experiment.

طلبہ کے ایک گروپ کو فلیم ٹیسٹ (Flame Test) کے ذریعے دھاتی آئنز کی شناخت کرنے کا کام دیا گیا۔ دیے گئے تجربے کی بنیاد پر درج سوالات کے جواب دیں۔

a.	Which wire can be used in flame test to make a small loop?	فلیم ٹیسٹ میں چھوٹا سا لوپ بنانے کے لیے کون سی تار استعمال کی جا سکتی ہے؟	1x4
b.	Why must the wire loop be cleaned before each test?	ہر ٹیسٹ سے پہلے تار کے لوپ کو صاف کرنا کیوں ضروری ہوتا ہے؟	
c.	Which colour flame does Barium produce?	بیریم (Barium) کس رنگ کا شعلہ پیدا کرتا ہے؟	
d.	Which colour flame does Potassium produce?	پوٹاشیم (Potassium) کس رنگ کا شعلہ پیدا کرتا ہے؟	

Q.4 Answer the following questions.

درج ذیل سوالات کے جواب دیں۔

a.	Which method is used to Prepare pure copper sulphate crystal?	خالص کاپر سلفیٹ کے کرسٹل تیار کرنے کے لیے کون سا طریقہ استعمال کیا جاتا ہے؟	1x4
b.	What colour change indicates the presence of water using anhydrous copper sulphate.	بے آب (Anhydrous) کاپر سلفیٹ کے ذریعے پانی کی موجودگی کو کون سی رنگت کی تبدیلی ظاہر کرتی ہے؟	
c.	How can purity of water be tested in laboratory?	لیبارٹری میں پانی کے خالص پن کو کیسے جانچا جا سکتا ہے؟	
d.	How do impurities affect the boiling point of water?	امپیورٹیز پانی کے نقطۂ ابال (Boiling Point) پر کس طرح اثر انداز ہوتی ہیں؟	

CHATTAK ACADEMY, WAH CANTT

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Model Questions Paper Chemistry SSC (COMOSITE)
FOR ANNUAL EXAMINATION 2026 & ONWARDS
Practical Based Assessment (PBA)
Chemistry Curriculum 2022-23 and Scheme of Studies 2006

Total Marks: 20 Time: 2 hours

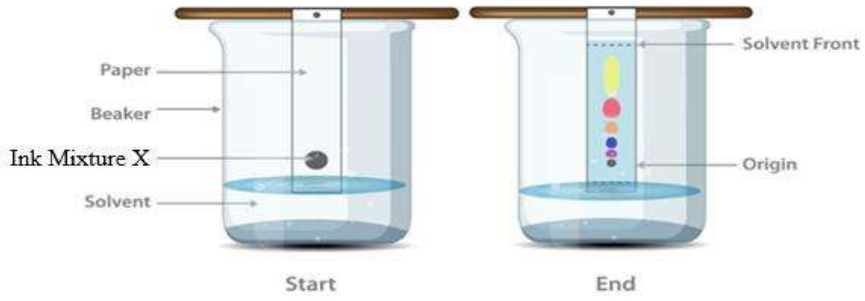
Note: Attempt all questions and write answers within provided spaces on E-Sheet.

SECTION A (6 x 2 = 12)

Q No.1 Ali, a Grade 10 student, is tasked with separating the components of a mixture of inks using paper chromatography in the school laboratory. He has been provided with the following materials:

- Chromatography paper
- A pencil
- A capillary tube
- Beakers
- A solvent (water/alcohol mixture)
- Ink samples (Mixture X)

Ali is instructed to follow the standard procedure for paper chromatography and analyze his results to identify the composition of mixture X.



a. Ali begins by drawing a baseline on the chromatography paper. Where should Ali draw the line, and why should he use a pencil instead of a pen? [1]

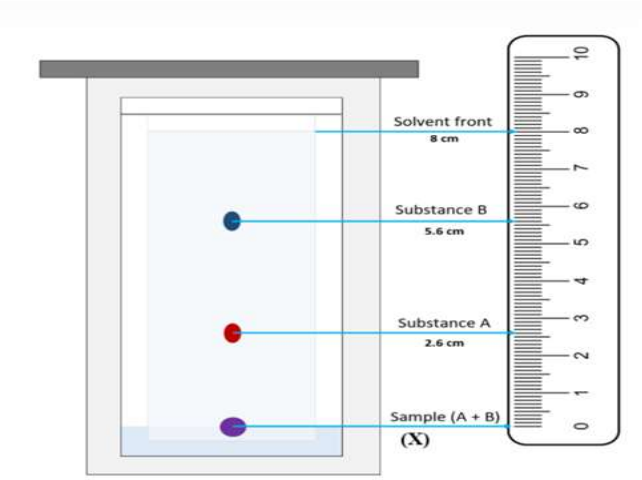
b. Ali used a capillary tube to apply small spots of X (mixture of inks). How can he ensure that the spots are properly placed for effective separation? [1]

c. After applying the ink spots, Ali placed the chromatography paper into a beaker with the solvent. What must Ali check to ensure the setup is correct before starting the experiment? [1]

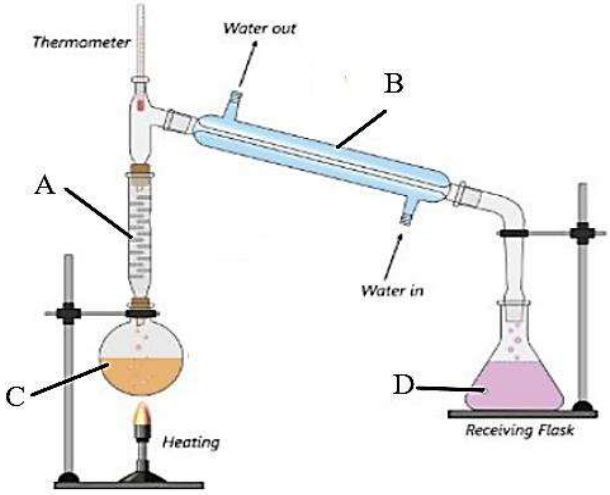
d. How can Ali determine when the experiment is complete, and what should he do immediately after removing the paper from the beaker? [1]

e. Ali observed that X separates into two distinct spots. What conclusion can Ali draw about the composition of X? [1]

f. Perform the necessary calculations to Calculate the Rf values for substance A and B. [1]



Q No.2 In the school laboratory, students are tasked with separating a mixture of water and alcohol using fractional distillation. The arrangement of the apparatus is shown in the figure below.



a. Label the components A, B, C and D. [1]

b. What is the purpose of the component labelled as `A` in this setup, and how does it improve the separation process? [1]

c. As you begin heating the mixture, which component (water or alcohol) will vaporize first, and why? [1]

d. During the experiment, the thermometer stabilizes at around 78°C. What does this indicate, and which liquid is being collected at this temperature? [1]

e. How can you confirm that the distillate collected at 78°C is alcohol and not water? [1]

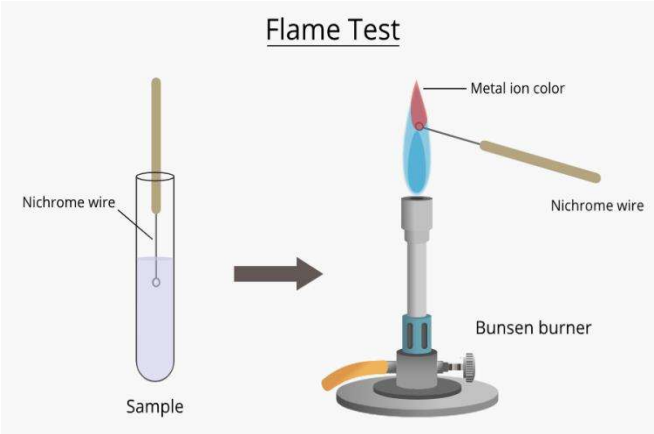
f. How would you adjust the heat source to ensure a steady and controlled distillation process? [1]

SECTION B (4x2 = 08)

Q No.3 A student **Y** is working in the school laboratory to identify metal ions using the flame test. He is provided with:

- Solid samples of salts containing Na^{+}, K^{+}, Ca^{2+},Cu^{2+}, and Ba^{2+} ions.
- A clean platinum/nichrome wire.
- Concentrated hydrochloric acid (for cleaning the wire).
- A Bunsen burner.
- Safety equipment (goggles, gloves).

Y must perform the flame test for each sample and record the flame color to identify the metal ion.



a. What is the purpose of cleaning the nichrome/platinum wire before testing a new sample. Why is this step important. [1]

b. Describe how **Y** should hold the wire in the Bunsen burner flame to observe the colour effectively. If **Y** observes a yellow flame while testing, which metal ion is likely present in the sample? [1]

c. A student, **Z** is setting up an experiment to measure the boiling point of ethanol. He repeats the experiment 3 times to calculate the average. His determined boiling points are 70ºC, 71ºC, 70.5ºC. The true boiling point of ethanol is 78ºC. How would you describe his work in terms of accuracy and precision? [1]

d. Design an experiment to separate a mixture of substances using sublimation. [1]

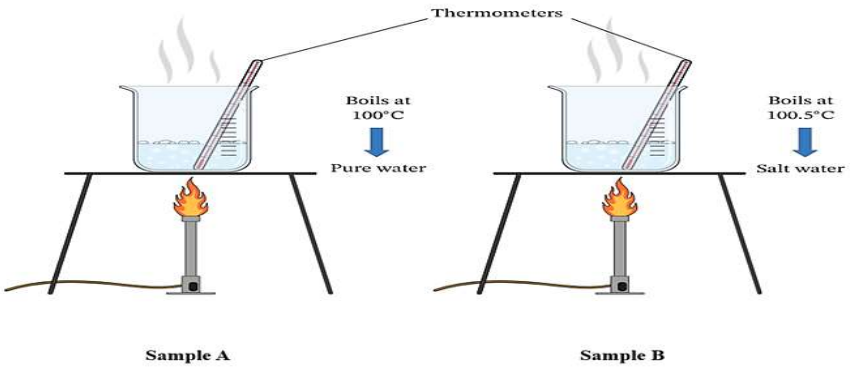
Q No.4

a. A group of students is conducting an experiment to test the purity of water by determining its melting and boiling points. They have two samples: **Sample A**, which they suspect is pure water, and **Sample B**, which they suspect contains impurities.

The students use a thermometer to measure the temperature at which both samples melt and boil.

They perform the following steps:

They boil both samples separately in an open container on a Bunsen burner and record the temperature at which the samples start to boil.



Here are the results:

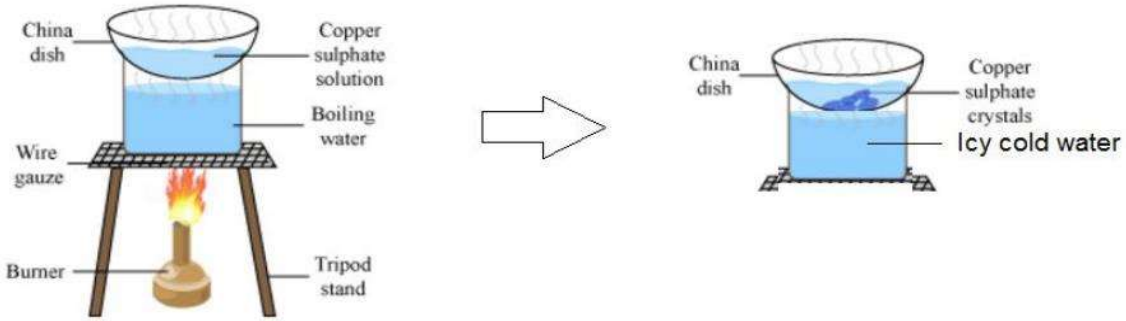
Sample A melts at 0°C and boils at 100°C.
Sample B melts at -2°C and boils at 102°C.

- Explain the significance of the results if pure water has a melting point of 0°C and a boiling point of 100°C at standard atmospheric pressure. [1]

- How can the experiment help determine whether a water sample is contaminated with substances like salt or other solutes? [1]

b. The metal displacement reactions involve the displacement of one metal ion from a solution by another metal. Students were provided with the zinc and copper solution to perform the experiment. What is the expected outcome of adding zinc metal to copper (II) sulphate solution? [1]

c. The image below shows crystals of Copper Sulphate forming during the cooling of the solution. Explain how the rate of cooling affects the size of the crystals. [1]



CHEMISTRY PBA

Class 10 / SSC-II • FBISE

Complete Questions & Solutions

1st Annual Examination 2026
Morning Group + Evening Group
and Chemistry PBA Model Paper

Reading order

Part 1: Morning Group - Questions 1 to 4
Part 2: Evening Group - Questions 1 to 4
Part 3: Model Paper - Questions 1 to 4

How to use this booklet

Read each question first and try answering it before looking below. Memorize the short exam-ready answer, then read the explanation to understand why it is correct.

All visible subparts are included in their original order. The supplied papers contain no OR alternatives. Calculations show the formula, substitution and final result.

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Study solutions, not an official FBISE marking scheme. Understand the concepts so you can answer questions with different wording.

Q1 | Lead and cadmium chromatography

Section A: Questions 1 and 2 total 12 marks. Separate Pb^{2+} and Cd^{2+} using paper chromatography. Materials: chromatography paper, capillary tube, covered glass jar, pencil and Na_2S locating agent.

(a) Why should a pencil be used instead of an ink pen to draw the baseline?

Answer: Pencil graphite does not dissolve in the solvent. Pen ink may dissolve and separate into coloured spots, interfering with the chromatogram.

(b) Which solvent or mixture of solvents should be used as the mobile phase?

Answer: One suitable mixture is **ethanol : 6 M nitric acid : distilled water = 8 : 1 : 1 by volume**.

Understand: This is a documented solvent system for Pb^{2+}/Cd^{2+} separation; follow your prescribed practical manual if its recipe differs. Na_2S is the locating agent, not the moving solvent.

(c) Which colour will be obtained for the Cd spot on the chromatogram?

Answer: A **yellow** spot forms because cadmium ions form cadmium sulphide.
$Cd^{2+}(aq) + S^{2-}(aq) \rightarrow CdS(s)$.

(d) Which colour will be obtained for the Pb spot on the chromatogram?

Answer: A **black** spot forms because lead ions form lead sulphide.
$Pb^{2+}(aq) + S^{2-}(aq) \rightarrow PbS(s)$.

(e) Calculate Rf for Pb and Cd: solvent front = 6.8 cm; Pb = 5.4 cm; Cd = 3.6 cm.

Answer: R_f = distance travelled by spot ÷ distance travelled by solvent front.
Pb: 5.4 ÷ 6.8 = 0.794 ≈ **0.79**.
Cd: 3.6 ÷ 6.8 = 0.529 ≈ **0.53**.

Understand: Rf has no unit. Measure from the same baseline to the centre of each spot. Pb travelled farther in this experiment.

Q2 | Identification of gases

Identify gases using damp litmus paper and limewater. Write the complete colour change, including the starting colour.

(a) What is damp litmus paper? Explain.

Answer: It is red or blue litmus paper moistened with distilled water, used to test acidic or alkaline behaviour.

(b) Why must litmus paper be damp while testing ammonia?

Answer: Ammonia dissolves in the water on the paper and produces hydroxide ions, making the solution alkaline.
$NH_3 + H_2O \rightleftharpoons NH_4^+ + OH^-$.

Understand: Water allows ammonia to show its alkaline behaviour.

(c) What colour change is observed in litmus paper while testing ammonia?

Answer: Ammonia turns **damp red litmus blue**.

(d) Write the equation when limewater reacts with carbon dioxide.

Answer: $Ca(OH)_2(aq) + CO_2(g) \rightarrow CaCO_3(s) + H_2O(l)$.

Understand: Calcium carbonate is the white precipitate that causes milkiness.

(e) How is carbon dioxide identified using limewater?

Answer: Bubble the gas through fresh limewater. Carbon dioxide turns it **milky** because a white calcium carbonate precipitate forms.

Understand: Excess CO_2 can remove the milkiness by forming soluble calcium hydrogen carbonate: $CaCO_3 + CO_2 + H_2O \rightarrow Ca(HCO_3)_2$.

(f) How is chlorine identified using damp litmus paper?

Answer: Chlorine turns **damp blue litmus red and then bleaches it white**.

Understand: $Cl_2 + H_2O \rightleftharpoons HCl + HClO$. The acidic solution causes reddening; hypochlorous acid causes bleaching.

Q3 | Boiling point of ethanol

Section B: Questions 3 and 4 total 8 marks. Students determine ethanol’s boiling point to assess its purity.

(a) What is meant by boiling point?

Answer: The temperature at which a liquid’s vapour pressure equals external pressure and the liquid boils throughout its bulk.

Understand: Evaporation occurs at the surface; boiling produces vapour bubbles throughout the liquid.

(b) What is the approximate boiling point of pure ethanol?

Answer: Approximately **78°C at standard atmospheric pressure (1 atm)**.

(c) What observation shows that ethanol has reached its boiling point?

Answer: Ethanol boils steadily, with vapour bubbles throughout the liquid, while the thermometer reading stays approximately constant near **78°C at 1 atm**.

Understand: During steady boiling, supplied heat changes liquid into vapour rather than raising its temperature. A few early bubbles alone may be escaping dissolved air.

(d) How does impurity affect the boiling point of ethanol?

Answer: A dissolved **non-volatile impurity generally raises the boiling point**. An impure liquid mixture may boil over a temperature range.

Understand: Volatile impurities can raise or lower the boiling temperature depending on their nature. Do not claim that every impurity always raises it.

Q4 | Sublimation and related concepts

Short practical questions: state the process, observation or effect directly.

(a) What is sublimation?

Answer: The direct change of a **solid into gas without passing through the liquid state**. Example: naphthalene.

Understand: The reverse change, gas directly into solid, is called deposition.

(b) How does impurity affect the melting point of a pure solid?

Answer: Soluble impurities generally **lower the melting point** and cause melting over a **wider temperature range**.

Understand: Remember both effects: lower melting point and broader melting range.

(c) What colour flame does sodium produce?

Answer: An intense **golden-yellow flame**.

(d) What happens to sand and salt during sublimation of naphthalene mixed with sand and salt?

Answer: Sand and common salt do not sublime under the conditions used, so they remain in the dish. Naphthalene sublimes and deposits as crystals on a cooler surface.

Understand: This separates naphthalene from the mixture. Sand and salt remain mixed together.

PART 2 • EVENING GROUP

Q1 | Determination of NaOH molarity

Section A totals 12 marks. Determine NaOH molarity using 0.1 M HCl. HCl is in the burette; 10 cm^3 NaOH is in the conical flask.

(a) What is volumetric analysis?

Answer: Determining an unknown solution’s concentration by measuring the volume of a standard solution required to react completely with it.

(b) Which indicator can be used for this analysis?

Answer: Phenolphthalein.

(c) What is the endpoint with the indicator in part (b)?

Answer: The **pink colour just disappears** and the solution remains colourless on swirling.

Understand: Acid is added to alkali here: pink → colourless.

(d) Write the reaction involved in the titration.

Answer: $HCl(aq) + NaOH(aq) \rightarrow NaCl(aq) + H_2O(l)$.
The mole ratio is 1 : 1.

(e) Calculate the mean volume V_2 of HCl used. Initial/final readings (cm^3): 0/9; 9/18; 18/27.

Answer: Volume used = final reading − initial reading.
Trial 1: 9 − 0 = **9 cm^3**.
Trial 2: 18 − 9 = **9 cm^3**.
Trial 3: 27 − 18 = **9 cm^3**.
Mean V_2 = (9 + 9 + 9) ÷ 3 = **9 cm^3**.

(f) Calculate NaOH molarity using V_1 = 10 cm^3 and HCl molarity M_2 = 0.1 M.

Answer: For this 1 : 1 reaction: $M_1V_1 = M_2V_2$.
$M_1 \times 10 = 0.1 \times 9$.
$M_1 = (0.1 \times 9) \div 10$ = **0.09 mol dm^{-3} = 0.09 M**.

Understand: Both volumes use cm^3, so they can be used directly in this ratio.

Q2 | Separation of ink dyes

Separate inks by paper chromatography. Measure every distance from the baseline.

(a) What is the stationary phase in paper chromatography?

Answer: The phase that remains fixed. In ordinary partition paper chromatography, it is **water held in the paper fibres**; the paper supports it.

Understand: School-level answers often call the chromatography paper the stationary phase. Water held in it is the more precise description for this method.

(b) Write the formula for Rf.

Answer: R_f = distance travelled by dye from baseline ÷ distance travelled by solvent front from baseline.

(c) What is the mobile phase in paper chromatography?

Answer: The solvent or solvent mixture that moves through the paper and carries the ink components with it.

(d) What is a chromatogram?

Answer: The pattern of separated spots or bands obtained after chromatography.

Understand: Chromatography is the process; a chromatogram is the resulting pattern.

(e) Calculate Rf: solvent = 8 cm; blue spot = 3 cm; red spot = 5 cm.

Answer: Blue: $R_f = 3 \div 8 =$ **0.375**.
Red: $R_f = 5 \div 8 =$ **0.625**.

Understand: Rf has no unit. The red dye travelled farther in this experiment.

Q3 | Flame tests

Section B totals 8 marks. Flame colours help identify metal ions.

(a) Which wire can be used to make a small loop for a flame test?

Answer: Platinum wire. Nichrome wire is a common alternative.

(b) Why must the wire loop be cleaned before each test?

Answer: To remove previous sample residues that could produce interfering flame colours and cause incorrect identification.

Understand: Clean the loop with hydrochloric acid and heat until it gives no characteristic flame colour.

(c) Which colour flame does barium produce?

Answer: An **apple-green flame**.

(d) Which colour flame does potassium produce?

Answer: A **lilac or pale violet flame**.

Q4 | Crystallization and water tests

Distinguish the presence of water from its purity.

(a) Which method is used to prepare pure copper sulphate crystals?

Answer: Crystallization, or recrystallization when purifying an impure sample.

Understand: Dissolve in minimum hot water, filter insoluble impurities, concentrate, cool, then filter and dry the crystals. Do not evaporate completely to dryness.

(b) What colour change indicates water using anhydrous copper sulphate?

Answer: White anhydrous copper sulphate turns blue.
$CuSO_4 + 5H_2O \rightarrow CuSO_4 \cdot 5H_2O$.

Understand: This confirms water is present; it does not establish purity.

(c) How can the purity of water be tested in the laboratory?

Answer: Measure its boiling point. Pure water boils sharply at **100°C at 1 atm**. Its freezing/melting point is approximately **0°C** at this pressure.

Understand: Compare at the same pressure: lower atmospheric pressure lowers even pure water's boiling point.

(d) How do impurities affect the boiling point of water?

Answer: Dissolved non-volatile impurities, such as salt, generally **raise its boiling point**.

Understand: They lower water's vapour pressure, so a higher temperature is needed for boiling at the same pressure. Volatile impurities may behave differently.

Q1 | Paper chromatography

Section A totals 12 marks. Ali separates ink mixture X using chromatography paper, pencil, capillary tube, beakers and a water/alcohol solvent.

(a) Where should Ali draw the baseline, and why should he use a pencil instead of a pen?

Answer: Draw it about **1–2 cm above the bottom edge**, above the solvent level. Pencil graphite is insoluble; pen ink may dissolve and interfere.

(b) How can Ali ensure that the small spots of X are properly placed for effective separation?

Answer: Put a small, concentrated spot on the baseline with the capillary tube. Let it dry before adding more at the same position. Keep separate sample spots well apart.

Understand: Large spots spread and may overlap, reducing separation clarity.

(c) What must Ali check before starting the experiment?

Answer: The paper's lower end must dip into the solvent, while the **baseline and ink spots stay above it**. Hang the paper vertically without touching the beaker sides.

(d) How does Ali know when to stop, and what should he do immediately after removing the paper?

Answer: Stop when the solvent front approaches the top, before it reaches the edge. Remove the paper and **immediately mark the solvent front in pencil**, then dry it.

Understand: The solvent front becomes invisible as the solvent evaporates.

(e) X separates into two distinct spots. What can Ali conclude about its composition?

Answer: X is a mixture containing **at least two dye components**.

(f) Calculate Rf for A and B. The diagram gives solvent front = 8 cm, A = 2.6 cm and B = 5.6 cm.

Answer: R_f = spot distance ÷ solvent-front distance.
A: 2.6 ÷ 8 = **0.325**.
B: 5.6 ÷ 8 = **0.70**.
Rf has no unit.

Q2 | Fractional distillation

Students separate water and ethanol. In the supplied drawing, A points to the column, B to the condenser, C to the starting liquid and D to the collected liquid.

(a) Label A, B, C and D.

Answer: A: Fractionating column.
B: Liebig condenser.
C: Water–alcohol mixture in the distillation flask.
D: Collected distillate, initially alcohol-rich, in the receiving flask.

(b) What is the purpose of A, and how does it improve separation?

Answer: It allows **repeated condensation and vaporization**. Rising vapour becomes richer in the more volatile ethanol, improving separation.

(c) Which component vaporizes first, water or alcohol, and why?

Answer: Ethanol vaporizes preferentially because it boils at about **78°C**, lower than water's **100°C** at 1 atm.

Understand: Both can enter the vapour, but the initial vapour is richer in ethanol.

(d) The thermometer stabilizes near 78°C. What does this indicate, and which liquid is collected?

Answer: The **ethanol-rich fraction** is distilling. The liquid collected is mainly ethanol.

Understand: Ordinary fractional distillation does not necessarily produce completely water-free ethanol.

(e) How can the distillate collected at 78°C be confirmed as alcohol rather than water?

Answer: In a separate supervised test, an ethanol-rich sample burns with a **pale blue flame**, whereas water does not burn.

Understand: Keep any flammability test away from the distillation apparatus and alcohol supply.

(f) How should the heat source be adjusted for steady, controlled distillation?

Answer: Maintain **gentle, steady boiling** so distillate collects slowly, drop by drop. Avoid vigorous boiling.

Understand: Use suitable controlled electric heating or a water bath for ethanol mixtures instead of an open flame.

Q3 | Flame tests and practical skills

Section B totals 8 marks. Salt samples contain Na^+, K^+, Ca^{2+}, Cu^{2+} and Ba^{2+} ions. A platinum/nichrome wire, hydrochloric acid and Bunsen burner are provided.

(a) Why must the wire be cleaned before a new sample, and why is this important?

Answer: To remove previous sample residues, preventing interfering flame colours and incorrect identification.

(b) How should the wire be held in the flame? Which ion is likely present if a yellow flame is observed?

Answer: Hold the sample-bearing loop in the hot outer region of a **non-luminous blue flame**, just above the inner blue cone's tip. Golden yellow indicates **sodium ions, Na^+**.

(c) Ethanol boiling points are 70°C, 71°C and 70.5°C, while the true value is 78°C. Describe accuracy and precision.

Answer: The results are **precise but inaccurate**: close together but far from 78°C.
Mean = (70 + 71 + 70.5) ÷ 3 = **70.5°C**.
The mean is **7.5°C below** the accepted value.

Understand: Precision: closeness of repeated results to one another. Accuracy: closeness to the accepted value.

(d) Design an experiment to separate a mixture using sublimation.

Answer: Use ammonium chloride mixed with sand. Place it in an evaporating dish under an inverted funnel with a loose cotton plug in its stem. Heat gently. Ammonium chloride sublimes and deposits on the cooler funnel surface; sand remains in the dish.

Understand: Use ventilation and teacher supervision. Collect the deposit after cooling. The method separates a sublimable solid from a non-sublimable substance.

Flame-colour revision

Na^+: golden yellow • K^+: lilac • Ca^{2+}: brick/orange-red
Cu^{2+}: blue-green • Ba^{2+}: apple green

Q4 | Purity, displacement and crystals

Written results: Sample A melts at 0°C and boils at 100°C; Sample B melts at −2°C and boils at 102°C. Assume standard atmospheric pressure.

(a)(i) Explain the significance of the results using pure water's melting and boiling points.

Answer: A is **consistent with pure water** because its values match 0°C and 100°C at 1 atm. B is **impure**: its melting point is lower and boiling point higher.

Understand: The illustration labels B as 100.5°C, but the written results state 102°C. Use the written 102°C; either supports the same qualitative conclusion.

(a)(ii) How does this experiment help detect salt or other solutes in water?

Answer: Compare melting/freezing and boiling points with pure water at the same pressure. Dissolved non-volatile solutes generally **lower the freezing/melting point and raise the boiling point**.

Understand: This indicates dissolved solutes but does not identify the particular substance.

(b) What is the expected outcome of adding zinc metal to copper(II) sulphate solution?

Answer: Zinc displaces copper because it is more reactive. **Reddish-brown copper** deposits, zinc dissolves and the blue solution fades as colourless zinc sulphate forms. **$Zn(s) + CuSO_4(aq) \rightarrow ZnSO_4(aq) + Cu(s)$**.

Understand: Blue colour can disappear if sufficient zinc reacts. Ionic equation: $Zn + Cu^{2+} \rightarrow Zn^{2+} + Cu$.

(c) How does the cooling rate affect copper sulphate crystal size?

Answer: Slow cooling: generally fewer, larger, well-shaped crystals.
Rapid cooling: generally many smaller crystals.

Understand: Slow cooling allows more time for particles to arrange on growing crystals. Rapid cooling creates many crystallization centres.

Exam checklist

State the observation and its reason. Balance equations. For calculations, show formula → substitution → result. Include units where needed; Rf is unitless.

Sources: uploaded Morning and Evening question images and Chemistry PBA Model Paper. Morning Q1(b) follows the Pb^{2+}/Cd^{2+} solvent system in NCERT Chemistry Laboratory Manual, Unit 5, Experiment 5.2.

Reference: NCERT chromatography practical manual